

FDE Xlerate

Week 1, Day 5

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The problem

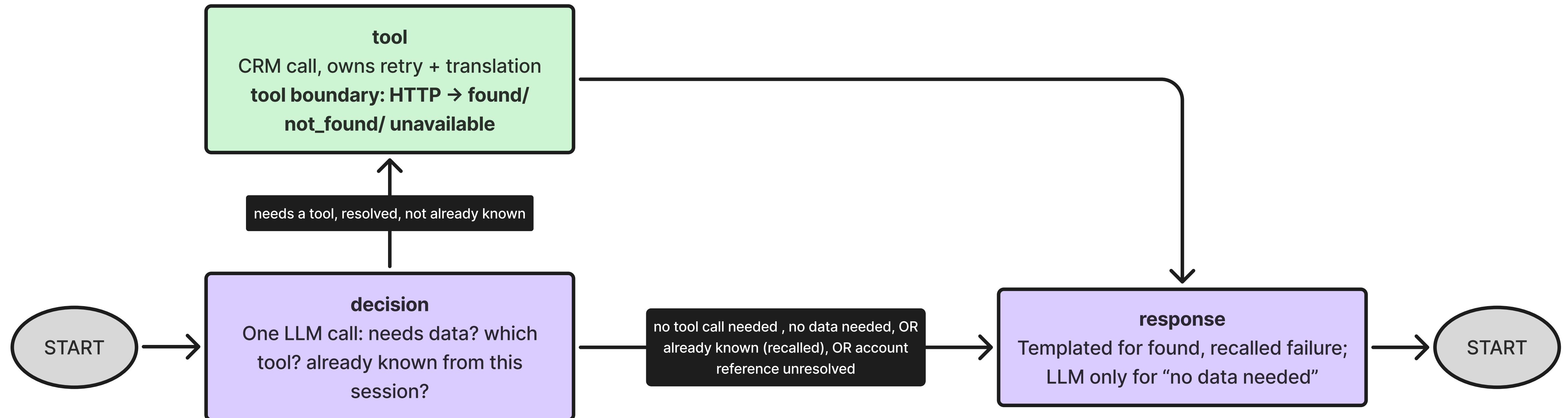
■ Task

- Finalize the PoC from day 4, with four bars to clear
 - Unit tests over the tool-calling logic specifically
 - Call/no-call, argument construction, every failure class
 - A demo that runs end-to-end from a clean checkout, no fixture fiddling between steps, including one deliberate failure
 - A peer review with a finding actually acted on before submission (N/A)
 - Every path the demo doesn't cover named as a known gap

■ Business problem

- Industry: banking
- Case: a customer-service agent that answers bank-account questions by calling a CRM
- Priorities
 - Minimize calling the CRM, ensuring it's only queried when there is requisite information the agent doesn't already have
 - The agent must never invent data. If querying the CRM returns nothing, the agent must say so plainly and offer escalation

High-level flow



Architecture - added from day 4

Tests

- **test_response.py**
- `compose_reply` had zero direct unit tests before; only 2 of its ~7 branches were reachable through the existing graph-integration tests.
- Now, all tested directly:
 - The `not_found` branch
 - All five `_compose_found` tool-formatters
 - The `recalled_outcome` branch
 - The LLM-stubbed `_compose_direct_reply` path
- 76 tests total, still under 3 seconds

Rehearsed Demo Script

- **demo_script.md**, **run.py**, and the **mock CRM** turned into a runbook:
 - Exact commands
 - Narration tied to specific code
 - A troubleshooting section
 - A live port-8000 conflict while rehearsing, now a documented pre-flight check

Known gaps

- **known_gaps.md**
- Outlines what the live demo won't show and why

Tech Stack

■ Mock CRM - unchanged

FastAPI + uvicorn, fixture data, header-driven fault override plus fixture-flagged demo accounts. Contract in mock_crm/contract.md

■ Orchestration - unchanged

LangGraph StateGraph (langgraph==1.2.11, same pin as day 3), with InMemorySaver checkpointing

■ Tests - day 5 focus

- LLM fully stubbed via dependency-injected client parameters (FakeMessagesClient)
- Same pattern reused across decision.py and response.py
- No network calls, deterministic, fast enough to run live in review
- Tool-layer tests run against a real uvicorn instance, not an in-process stub

■ Tool layer - unchanged

httpx, explicit timeouts, hand-rolled retry (3x, exponential backoff + jitter)

■ Model/ SDK - unchanged

claude-sonnet-5 via anthropic==0.104.1, same pin carried from days 2–3